

PHASE 4-PROJECT PLANNING

Project Title: POCKETSMART AI

Your Smart Budget & Recommendation Assistant

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1. Project Overview

PocketSmart AI is an intelligent budget management and recommendation assistant designed to help users manage their personal expenses effectively. The system allows users to record income and expenses, categorize spending, monitor budgets, and receive personalized recommendations based on their spending patterns. By using Artificial Intelligence, the application can provide useful suggestions to improve financial planning and encourage responsible spending.

2. Problem Statement

Many people find it difficult to track their daily expenses and understand where their money is being spent. Manual budgeting can be time-consuming, and users may not notice unnecessary spending until the end of the month. Existing expense trackers may record transactions but do not always provide personalized recommendations. PocketSmart AI aims to solve this problem by combining budget tracking with intelligent, user-friendly recommendations.

3. Objectives

- To develop a simple and user-friendly personal budget management system.
- To allow users to record, categorize, and monitor income and expenses.
- To help users set and track monthly or category-wise budgets.
- To analyze spending patterns and identify areas of high expenditure.
- To provide personalized recommendations based on the user's financial activity.
- To display useful summaries and reports for better financial planning.

4. Proposed System

The proposed system provides a centralized platform for managing personal finances. Users can enter their income and expenses, assign categories, set budget limits, and view their spending summaries. The AI recommendation component analyzes the available spending data and generates suggestions, such as reducing excessive spending in a particular category or adjusting a budget based on previous spending patterns.

5. Key Features

- User registration and secure login.
- Income and expense management.
- Expense categorization.
- Monthly and category-wise budget setting.
- Dashboard showing income, expenses, balance, and budget status.
- Spending analysis and visual reports.
- AI-based personalized recommendations.
- Budget alerts and spending notifications.
- Transaction history and summary reports.

6. Project Planning

The project will be developed through a sequence of planning, design, development, testing, and deployment activities. Each phase will be completed systematically to ensure that the final application meets the defined requirements.

Phase	Activity	Expected Output	Duration
1	Requirement Analysis	Functional and non-functional requirements	1 Week

2	System Design	Architecture, database and UI design	1 Week
3	Development	Working application modules	3 Weeks
4	AI Recommendation Module	Recommendation and analysis features	1 Week
5	Testing	Tested and corrected application	1 Week
6	Documentation & Deployment	Final report and deployable system	1 Week

7. Technology Requirements

- Frontend: HTML, CSS, JavaScript / React (as applicable)
- Backend: Python with Flask or Django (as applicable)
- Database: MySQL / SQLite
- AI/ML: Python-based recommendation or analysis module
- Development Environment: Visual Studio Code or equivalent IDE
- Version Control: Git/GitHub

8. Functional Requirements

- The system shall allow users to create and manage their accounts.
- The system shall allow users to add, edit, and delete income and expense records.
- The system shall categorize transactions.
- The system shall allow users to set budget limits.
- The system shall calculate spending and remaining budget automatically.
- The system shall analyze transaction data and generate recommendations.
- The system shall display financial summaries and reports.

9. Non-Functional Requirements

- Usability: The interface should be simple and easy to understand.
- Performance: The system should process normal user transactions quickly.
- Security: User information and financial records should be protected.

- **Reliability:** The application should provide consistent results.
- **Scalability:** The system should support additional features in the future.
- **Maintainability:** The software should be organized for easy updates and maintenance.

10. Expected Outcome

The expected outcome of PocketSmart AI is a practical budget management application that helps users understand their spending habits and make more informed financial decisions. The AI-based recommendation feature will add value by converting transaction data into simple, personalized suggestions.

11. Conclusion

The planning phase establishes the requirements, objectives, features, technologies, and development schedule for PocketSmart AI. A structured planning approach will help ensure that the project is developed efficiently and that the final system provides an effective combination of budgeting, expense tracking, analysis, and personalized recommendations.